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Publications

20. J. Bruce, L. Cranton Heller, M. Sayrafi. Characterizing Multigraded Regularity and Virtual Resolutions on Products of Projective Spaces. *Advances in Mathematics*. E-Print: [arXiv:2110.10705](#)
19. J. Bruce. Asymptotic syzygies for products of projective spaces. *Journal of Algebra*, **649** (2024), 347-391. E-Print: [arXiv:1904.04944](#)
18. M. Brandt, J. Bruce, M. Chan, M. Melo, G. Moreland, C. Wolfe. On the Top-weight Cohomology of $\mathcal{A}_g$. *Geometry & Topology*, **28**:2 (2024), 497-538. E-Print: [arXiv:2012.02892](#)
17. J. Bruce, D. Corey, D. Erman, S. Goldstein, R. Laudone, and J. Yang. Syzygies of $\mathbb{P}^1 \times \mathbb{P}^1$: Data and Conjectures. *Journal of Algebra*, **593** (2022) no. 1, 589-621. E-Print: [arXiv:2104.14598](#)
16. J. Bruce, D. Erman, S. Goldstein, and J. Yang. The Schur-Veronese package in Macaulay2. *Journal of Software for Algebra and Geometry*, **11** (2021), 83-87 E-print: [arXiv:1905.12661](#).
15. J. Bruce. The Quantitative Behavior of Asymptotic Syzygies for Hirzebruch Surfaces. *Journal of Commutative Algebra*, **14** (2022) no. 1, 19-26. E-Print: [arXiv:1906.07333](#).
14. A. Almousa, J. Bruce, M. Loper, and M. Sayrafi. The Virtual Resolutions Package for Macaulay2. *Journal of Software for Algebra and Geometry*, **10** (2020), 50-60. E-print: [arXiv:1905.07022](#).
13. J. Bruce and D. Erman. A probabilistic approach to systems of parameters and Noether normalization. *Algebra and Number Theory*, **13** (2019), no. 9, 2081–2102. E-print: [arXiv:1604.01704](#).
12. J. Bruce and W. Li. Effective bounds on the dimensions of Jacobians covering abelian varieties. *Proc. Amer. Math. Soc.*, **148** (2020), no. 2, 535-551. E-print: [arXiv:1804.11015](#).
11. J. Bruce, D. Erman, S. Goldstein, and J. Yang. Conjectures and computations about Veronese syzygies. *Experimental Mathematics*, **29** (2020), 398-413. E-print: [arXiv:1711.03513](#).
10. M. Brandt, J. Bruce, T. Brysiewicz, R. Krone, and E. Robeva. The degree of $SO(n)$. *Combinatorial Algebraic Geometry*, 207-224, Fields Inst. Commun. **80** (2017). E-print: [arXiv:1701.03200](#).
9. J. Bruce, M. Logue, and R. Walker. Monomial valuations, cusp singularities, and continued fractions. *Journal of Commutative Algebra*, **7** (2015) no. 4, 495-522. E-print: [arXiv:1311.6493](#).
8. J. Bruce, P. Kao, E. Nash, B. Perez, and P. Vermeire. Betti tables of reducible algebraic curves. *Proc. Amer. Math. Soc.* **142** (2014) 4039-4051. E-print: [arXiv:1210.3064](#).

Pre-Prints

7. E. Assaf, M. Brandt, J. Bruce, M. Chan, R. Vlad. Level structures on tropical abelian varieties. *Pre-print*. E-Print: [arXiv:2609.14895](#)
6. J. Bruce. Eventual Nonstandard Koszulness Fails for Veronese Subrings of Weighted Polynomial Rings. *Submitted*. E-Print: [arXiv:2608.23913](#)
5. J. Bruce, J. Bucciarelli, B. Zacovic. Explorations of Matroid Complexes. *Submitted*. E-Print: [arXiv:2605.24695](#)
4. J. Bruce, L. Cranton Heller, M. Sayrafi, A. Seceleanu. Seshadri Regions and the Asymptotic Shape of Multigraded Regularity. *Submitted*. E-Print: [arXiv:2512.05289](#)
3. E. Assaf, M. Brandt, J. Bruce, M. Chan, R. Vlad. Tropicalizations of locally symmetric varieties. *Submitted*. E-Print: [arXiv:2505.10504](#)
2. M. Brandt, J. Bruce, D. Corey. The virtual Euler characteristic for binary matroids. *Submitted*. E-Print: [arXiv:2301.10108](#)
1. J. Bruce, L. Cranton Heller, M. Sayrafi. Bounds on Multigraded Regularity. *Submitted*. E-Print: [arXiv:2208.11115](#)

Non-Research Articles

2. J. Bruce. A Word from... Juliette Bruce, Inaugural President of Spectra. *Notices of the AMS*, **69** (2022) no. 6, 898–899.
1. A. Bonato, J. Bruce, and R. Buckmire. Spaces for All: The Rise of LGBTQ+ Mathematics Conferences. *Notices of the AMS*, **68** (2021) no. 6, 998–1003. <https://doi.org/10.1090/noti2288>